Arnav Revankar

Senior Computer Engineer @ NJIT

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Summary

Motivated Computer Engineering student with a passion for Embedded Systems. Experienced with STM32, ESP32, and RISC-V hardware/software development. Seeking Spring 2027 full-time roles

Education

New Jersey Institute of Technology (Sept 2023 - Present) — Cumulative GPA: 3.5

BS Computer Engineering — Newark, NJ

- **Coursework:** Circuits and Systems 1 & 2, Microprocessors, Computer Architecture, Applied Machine Learning, Digital Design, and others.
- **Activities:** IEEE-HKN Inductee, 2026 Hackathon Volunteer, NJIT Solar Car (Hardware + Software Team)

Technical Experience

Embedded Software/Hardware R&D Intern — July 2026 - Present

Excelitas — Billerica, MA

- Developing firmware from scratch for a TI C2000 microcontroller, running real time closed loop control algorithms for X-Ray and Cap charger high voltage power supplies.
- Contributing to next-gen hardware schematic design, to be built around the existing C2000 control card. Consists of mains power rectification, boost conversion with active PFC (via C2000), and more.
- Control algorithms currently driven by analog-to-discrete conversions via Bilinear transform, and 2P2Z controller implementations on the C2000's CLA.
- Working to meet MISRA C standards, and IEC 61000-3-2 for minimizing harmonic distortion.

Software Engineering Intern — March 2025 - August 2025

Mathnasium — Chatham, NJ

- Developed an email-based report generation and delivery software for a Mathnasium tutoring center.
- Currently being used to send 110-150 emails per day, saving 5 centers 30-40 hours a week.

Skills

- **Languages:** C, C++, MATLAB, Java, Kotlin, Python, Go, SQL (MySQL), GNU Octave, LaTeX, Javascript, VHDL
- **Software Tools:** Linux/Unix, KiCAD, FreeCAD, TI CCS & SysConfig, STM32CubeMX & IDE, PlatformIO, ESP-IDF, Git, Docker, FreeRTOS, SciPy
- **Build Tools/Debuggers:** Apache Maven, Apache Ant, CMake, Make, MSVC & GCC, GDB, Valgrind
- **Communication Protocols:** UART, I2C, SPI, CAN, USB 2.0, TCP and Socket programming in C/C++ and Go

Projects

8-Channel Logic Analyzer (Github) — Feb 2026 - May 2026

- USB Logic Analyzer built around an STM32F446, (theoretically) capable of sniffing upto 500kHz signals.
- DMA data acquisition, and run-length encoded data transmission, to optimize byte usage on USB 2.0 FS.
- UART, SPI, I2C, and CAN signal decoding via a custom Go backend.

BitSum Lecture Summarizer (GitHub) — Oct 2025 - Dec 2025

- Built a lecture summarizer tool, integrating Microsoft BitNet, OpenAI Whisper, and a custom feedforward network. Runs 2x faster than human comprehension on average.
- Integrated with Docker to ensure reproducibility.

SolarVision Autonomous Mower (Github) — Sept 2025 - May 2025

- YOLO and LiDAR-based autonomous lawnmower, engineering capstone project.
- Designed a custom 6-layer PCB for STM32N6 MCU, with external SPI NOR Flash and SDRAM Memory through the FMC.
- Trained a YOLOv8 Segmentation model on a custom dataset, and exported to C for the STM32N6 NPU, with the ST Edge AI Tool Suite.
- LiDAR data acquisition and path planning on an ESP32S3